

## Introduction

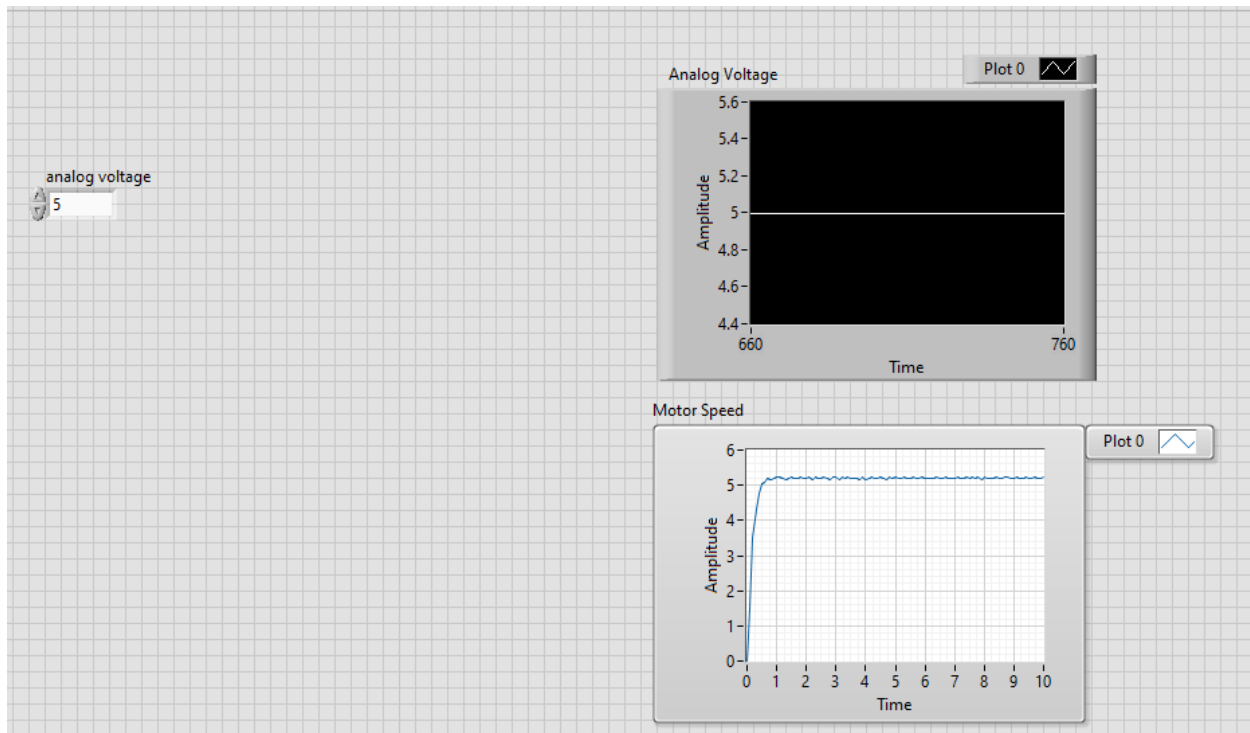
This lab utilized LabView to implement a PID controller to control a DC motor's speed. This was very similar to the Lab8a Motor modeling lab, and we reused the transfer functions we found in Lab 7. I plotted the tachometer's parameters (voltage and rpm) backwards in my Lab 7 Report, so I took the inverse of that and I was able to get a better transfer function.

Transfer function for tachometer =  $\frac{voltage}{rpm} = 0.0021$

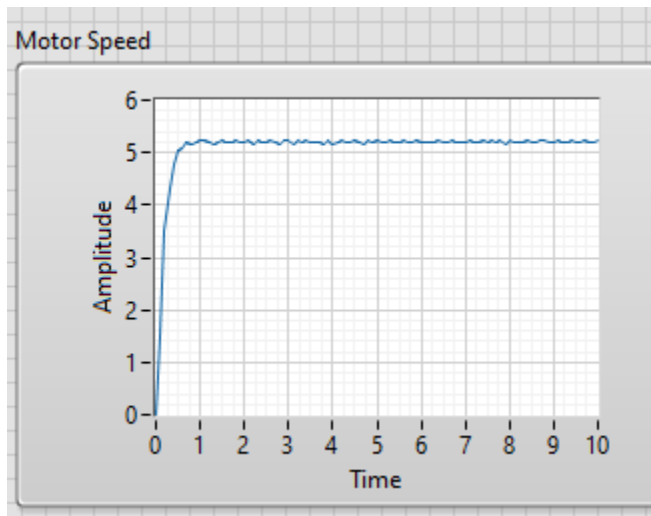
I was tasked with following given instructions to set up the block diagram, fairly similar to Lab 8a Redo, then adjusted the PID values to get the best response.

# Analysis

Front Panel:



Best Generated Response:



The system reaches a steady state in about half a second, is stable steadying around 5.2 V, has an overshoot of about 0.2 V, and is fairly smooth with minor oscillations.

$$\text{Percent Overshoot} = \text{PO} = \frac{y_{\max} - y_{ss}}{y_{ss}} 100\% = \frac{5.21 - 5.2}{5.2} 100\% = 0.19\%$$

$$\text{PO} = 0.19\%$$

This is the minimum / best PO.

Delay Time =  $t_d$  = time take system reach 50% of steady state output

$$y(t_d) = 5.2 / 2 = 2.6$$

$$t_d = 0.1 \text{ s}$$

Rise Time =  $t_s$  = time take system go from 10% to 90% steady state output

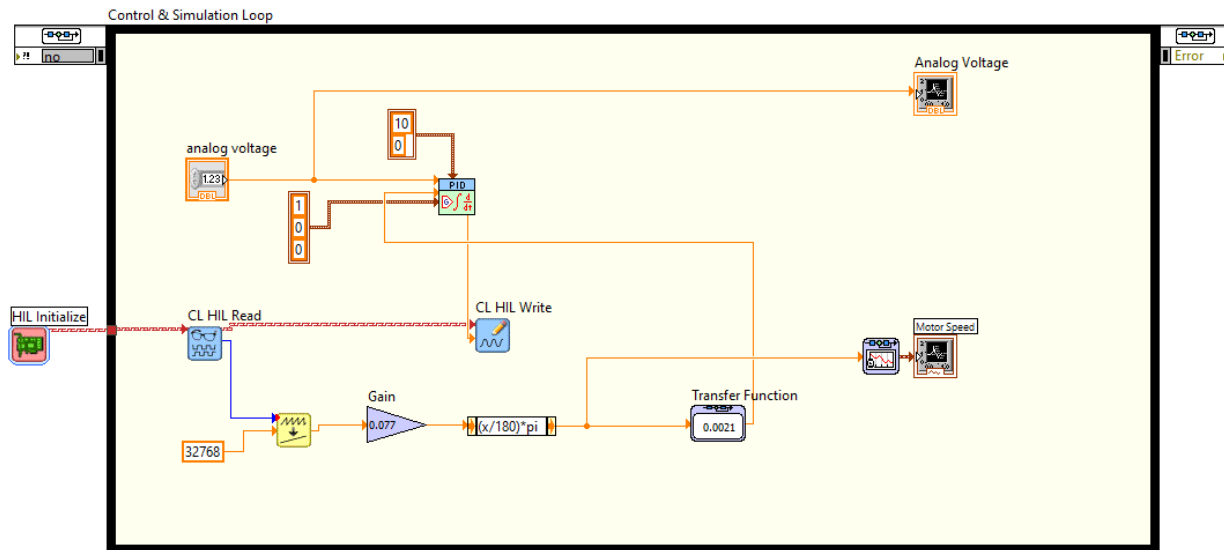
$$y(t_{10}) = 0.52$$

$$y(t_{90}) = 4.68$$

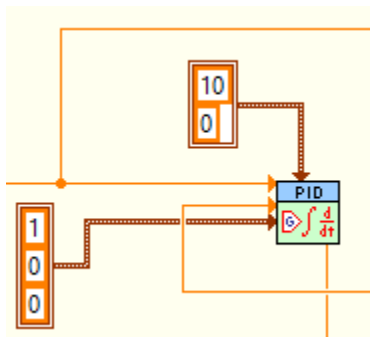
$$t_s = 0.35 - 0.05 = 0.3$$

$$t_s = 0.3$$

## Block Diagram:

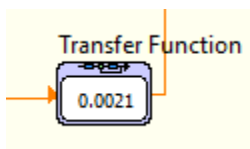


## PID:



My proportional constant  $K_p$  value was 1 with the integral and derivative constants equal to 0.

## Transfer Function:



## Conclusion

This lab was fairly straightforward. It was fun to play with the PID constants and see how those affected the output. This was a nice application of seeing what the PID constants actually do and will be useful knowledge later in the design course whenever we calculate the PID constants mathematically.